

Problem 1: DNN and Wine Classification (80 points)

<https://github.com/aidangoodman/ee446/tree/main/assignment%201>

(e) [Pro/Dis] Further Model Size Reduction (10 pts)

Question: Can you further reduce model size beyond parts (b), (c), and (d)? If yes, implement your proposed method and compare results. If not, explain why.

Answer: Yes I believe that we could further reduce the size by combining all the steps we did previously to the student model, mainly the quantitation and pruning steps. I basically copied the sections from earlier in the homework and applied the pruning steps first, then converted to a TFLite model for the int8 quantization. I was able to get the model size down all the way to 5.82 KB. I experimented to even decrease the sparsity more but the accuracy began to go down to 70% with only 0.05 KB decrease.

Problem 2: Exploring Edge Impulse (20 points)

1. [Dis] Learning Blocks (5 pts)

Question: What types of “Learning Blocks” are available under the free-tier Edge Impulse account (e.g., supervised learning—classification and regression, unsupervised learning—clustering, self-supervised learning, etc.)? For each type, provide at least one example model that can be implemented using the free tier.

Answer:

1. Supervised Learning Classification: - This block is able to learn patterns from data and apply them to new input data. It basically works by categorizing data into different groups, and can apply that categorization to new data inputs. This would be good for a model that tracks running, walking body movements, and can guess what a person is doing based on the data.
2. Supervised Learning Regression - This is similar to classification but without the categories. It rather intakes a running stream of data and can predict the output. This would be a good block for using in a weather forecasting model.
3. Unsupervised Learning Clustering (GMM and K-Means) - With clustering we do not need to have labels associated with any of our data points, we can push all the data into the clustering algorithm and it will decide where the buckets belong based on mathematical distance and difference. Different algorithms will work better and categorizing different data, such as K-Means for simple linear segments, or Gaussian for more overlapping data-points. This would be great for a model that has a lot of categorized data points, such as consumer recommendations where the model has a list of people with similar interests, which can then recommend you something you might also like.

4. Transfer Learning - These are great when we want to use a larger model that someone has already trained completely for us. We can use the feature and classification layers of the main model, and re-train the deeper layers to get our desired output. This is great for tiny-ml cases where a large more can accomplish something like image classification of car types, and we can retrain the model to work on boat types.

2. [Dis] Classifier Layers (5 pts)

Question: List all the types of layers that can be used under the “Classifier” section in the free-tier Edge Impulse. Briefly explain the purpose of each layer identified.

Answer:

1. Dense - This is the common type of neural network layer we have been using, is a fully connect neuron layer to the previous layer. We use it to process the features of input, with neurons corresponding to classes
2. 1D and 2D Convolution / Pooling - These layers are CNN and apply a filter over the data. We can use Convolution for audio signals, sequential data, and 2D image data. The pooling operations can shrink down the network output to make the overall processing quicker.
3. Reshape - This layer can change the dimensional space of our data with no effect on the content. This is useful for when we need to up-size certain data sets to larger inputs layers such as convolution or dense layers.
4. Flatten - This does the opposite of the reshape layer, but reduces the dimensional space of the data with no losses. It can take image data and convert it into an array of input for easier processing.
5. Dropout - Helps with overfitting by pruning network connections during training.

3. [Dis] Model Compression Options (5 pts)

Question: What types of model compression options are available under the “Deployment” section in the free-tier Edge Impulse? List all available methods.

Answer:

1. Quantized (int8)
2. Unoptimized (float32)

4. [Dis] Synthetic Data Generation (5 pts)

Question: What input modalities are supported for synthetic data generation in the free-tier version of Edge Impulse? For each modality, list the synthetic data generation methods/algorithms provided. Briefly explain why synthetic data is important in machine learning applications.

Answer:

1. GPT-Image-2 Synthetic Image Generator / Black Forest Labs Flux-Pro Synthetic Image : We have two image generating models for synthetic data creation.
2. Whisper Synthetic Voice Generator: We also have a text to speech model to generate audio data.

These models are important for machine learning since it allows us to quickly and effectively test high quality data without the need to spend a lot of money and time collecting external data. It almost acts as a sort of knowledge distillation to other models, and provides large datasets, and good labels. Moreover, people don't have to worry about their own data being used to train models, which raises a lot of privacy concerns nowadays